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Chapter 5

Basic Router Configuration

The Cisco CCNA certification exams — in particular, the ICND1 exam — will test you on the basic configuration of a Cisco router. This chapter is designed to test you on the basics of router configuration, including interface configuration, global configuration commands, and basic troubleshooting of the interfaces. Be very familiar with the configuration commands, and also make note of the troubleshooting scenarios that are presented in this chapter (or any chapter): They are the types of questions you can expect on the CCNA exams!

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Viewing and altering the command history
- Understanding basic router configuration settings
- Configuring Fast Ethernet and serial interfaces
- Configuring different types of passwords on the Cisco router
- Troubleshooting interface problems

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Know basic configuration commands such as changing the console time-out, history size, and the hostname of the router.
- Be aware of the prompt at which you are typing each command.
- You can erase the configuration of a device by erasing the startup-config and then reloading the router.
- Know your show commands, such as `show interfaces`. The `show interfaces` command and the `show ip interface brief` command are used to display the status of the interfaces, whereas the `show controllers` command can be used to view whether a serial interface is a DTE or DCE device. Specifically know how to view the status of an interface and know the meaning of an interface status such as "Serial0/2/0 is down, line protocol is down".

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Configuring Interfaces

162–172 Express the given number in scientific notation.

162. You wish to disable the Fast Ethernet interface on your router. What command would you use?

(A) router(config-if)#`shutdown`

(B) router(config-if)#`disable`

(C) router(config)#`shutdown interface F0/0`

(D) router(config)#`disable interface F0/0`

163. Which of the following are encapsulation protocols that can be used on a serial link for a Cisco router? (Select all that apply.)

(A) IPX

(B) PPP

(C) PPTP

(D) HDLC

(E) RIP

164. You would like to configure the Fast Ethernet port on your router for full duplex. What commands would you use?

(A) Use the following commands:

R1#`config term`

R1(config)#`interface f0/0`

R1(config-if)#`duplex full`

(B) Use the following commands:

R1#`config term`

R1(config)#`interface f0/0`

R1(config-if)#`duplex half`

(C) Use the following commands:

R1#`config term`

R1(config)#`interface f0/0`

R1(config-if)#`set full duplex`

(D) Use the following commands:

R1#`config term`

R1(config)#`interface f0/0`

R1(config-if)#`set half duplex`

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165. You wish to enable the Fast Ethernet port on the router. What command would you use?

(A) R1(config-if)#`undo shutdown`

(B) R1(config-if)# `enable`

(C) R1(config-if)#`no shutdown`

(D) R1(config-if)#`no disable`

166. Which of the following layer-2 protocols are supported encapsulation protocols on the serial link of a Cisco device? (Select two.)

(A) PPP

(B) RIP

(C) OSPF

(D) HDLC

167. Which of the following commands would you use to assign an IP address to the Fast Ethernet interface on the router?

(A) router(config-if)#`ip address 10.0.0.1`

(B) router(config)#`ip address 10.0.0.1 255.0.0.0`

(C) router(config-if)#`ip address 10.0.0.1 255.0.0.0`

(D) router(config)#`ip address 10.0.0.1`

168. When looking at the interface fastethernet 0/1 command, what is the port number for the interface?

(A) 0

(B) 1

(C) 2

(D) 3

169. In a point-to-point serial link between two routers, which router sets the clock rate?

(A) AUI

(B) DTE

(C) BNC

(D) DCE

170. Which of the following are additional commands needed to configure a serial interface over what is needed when configuring an Ethernet interface? You are configuring the DCE device on a serial link. (Select all that apply.)

(A) `clock rate 64000`

(B) `description WAN link`

(C) `encapsulation hdlc`

(D) `ip address dhcp`

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171. After disabling the serial interface using the shutdown command, which of the following represents the output displayed on the console as a result of your action?

(A) serial0/0 is administratively down, line protocol is administratively down

(B) serial0/0 is down, line protocol is down

(C) serial0/0 is administratively down, line protocol is down

(D) serial0/0 is down, line protocol is administratively down

172. What command do you use on the interface to assign an IP address to the interface automatically via DHCP?

(A) R1(config)#`ip address dhcp`

(B) R1(config)#`ip dhcp address`

(C) R1(config-if)#`ip dhcp address`

(D) R1(config-if)#`ip address dhcp`

Basic Configuration

173–186 Express the given number in scientific notation.

173. Which of the following commands would you use to change the hostname on the router?

(A) router>`hostname R1`

(B) router(config-if)#`hostname R1`

(C) router#`hostname R1`

(D) router(config)#`hostname R1`

174. Which of the following password types are encrypted in the configuration by default?

(A) enable password

(B) enable secure password

(C) enable secret

(D) enable encrypted password

175. You are at the following prompt. What command would you type first to change the name of the router?

Router (config-if)#

(A) `hostname`

(B) `exit`

(C) `disable`

(D) `enable`

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176. Which of the following command-sets would you use to configure a password on the console port of the router?

(A) Use the following commands:

```
enable
password myConPass
login
```

(B) Use the following commands:

```
enable
config term
line con 0
password myConPass
login
```

(C) Use the following commands:

```
enable
config term
line con 0
password myConPass
```

(D) Use the following commands:

```
enable
config term
password myConPass
login
```

177. You are looking to create a username for Tom to log on to the router. What command would you use?

(A) r1(config)# `username tom password P@ssw0rd`

(B) r1# `username tom password P@ssw0rd`

(C) r1> `username tom password P@ssw0rd`

(D) r1(config-if)# `username tom password P@ssw0rd`

178. You wish to view a list of commands recently used on the router, what command would you use?

(A) `show history`

(B) `show commands`

(C) `show all`

(D) `show my commands`

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179. You would like to encrypt all passwords configured on the router. What command would you use?

(A) `enable password-encryption`

(B) `service password-encryption`

(C) `encrypt-password all`

(D) `encrypt-passwords`

180. You have made changes to your Cisco router and would like to verify the changes before saving them. What command would you use?

(A) `show startup-config`

(B) `show memory`

(C) `show running-config`

(D) `show flash`

181. You are administering a Cisco router. When you have a typo in the command name, you have to wait for the router to do a DNS lookup before getting control of the console again — an annoying problem. What command would fix it?

(A) `disable dns`

(B) `no ip domain lookup`

(C) `ip domain lookup`

(D) `no dns lookup`

182. You would like the console to time out after a minute and 45 seconds of inactivity. What command would you use?

(A) `exec-timeout 1 45`

(B) `exec-timeout 1:45`

(C) `no exec-timeout 1 45`

(D) `no exec-timeout 1:45`

183. What command can be used to modify the history size on the Cisco router?

(A) R1#`terminal history size 30`

(B) R1(config)#`terminal history size 30`

(C) R1(config)#`history size 30`

(D) R1#`history size 30`

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184. After creating a user named Bob, you would like to ensure that the console port prompts for username and password. What command would you use?

(A) r1(config-line)# `login`

(B) r1(config-line)# `enable login`

(C) r1(config-line)# `login local`

(D) r1(config-line)# `logon`

185. Which of the following banner types is the first to be displayed?

(A) Login

(B) MOTD

(C) Exec

(D) Logout

186. You would like to delete the configuration on your router and start with a clean configuration. Which two commands would you use? (Select two.)

(A) `delete startup-config`

(B) `reload`

(C) `restart`

(D) `erase startup-config`

(E) `format-router`

Basic Troubleshooting

187–199 Express the given number in scientific notation.

187. A network technician is having trouble connecting two routers across the serial ports. One router is a Cisco router, whereas the other router is from a different manufacturer. You have loaded HDLC on both routers. What should you do?

(A) Ensure the subnet mask on the serial ports is 255.255.255.252.

(B) Change the frame size on the serial interfaces.

(C) Use PPP as the serial link protocol.

(D) Change the buffer size on the serial interfaces.

188. You would like to verify the status of one of your interfaces. What command would you use?

(A) `show interfaces`

(B) `show ip route`

(C) `show router`

(D) `show f0/0 status`

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189. You are having trouble communicating with networks that are connected to your FastEthernet0/0 port. You use the command shown below to view the status of the links. What can you do to solve the problem?

```
R1>show ip interface brief
Interface              IP-Address   OK? Method Status
FastEthernet0/0        23.0.0.1     YES manual administratively down
FastEthernet0/1        unassigned   YES manual up
Serial0/2/0            24.0.0.1     YES manual up
Serial0/3/0            unassigned   YES manual administratively down
```

(A) Configure an encapsulation protocol.

(B) Assign an IP address.

(C) Configure it to use DHCP.

(D) Enable the interface.

190. Which of the following commands is used to display a table (output shown below) indicating the IP assigned to each interface and the status of the interface?

```
Interface          IP-Address   OK? Method Status
FastEthernet0/0    23.0.0.1     YES manual up
FastEthernet0/1    unassigned   YES manual up
Serial0/2/0        24.0.0.1     YES manual up
Serial0/3/0        unassigned   YES manual administratively down
```

(A) `show ip interfaces`

(B) `show ip table`

(C) `show ip interface brief`

(D) `show ip table brief`

191. Which of the following commands display information on the IOS version? (Select three.)

(A) `show version`

(B) `show IOS`

(C) `show hardware`

(D) `show flash`

(E) `show running-config`

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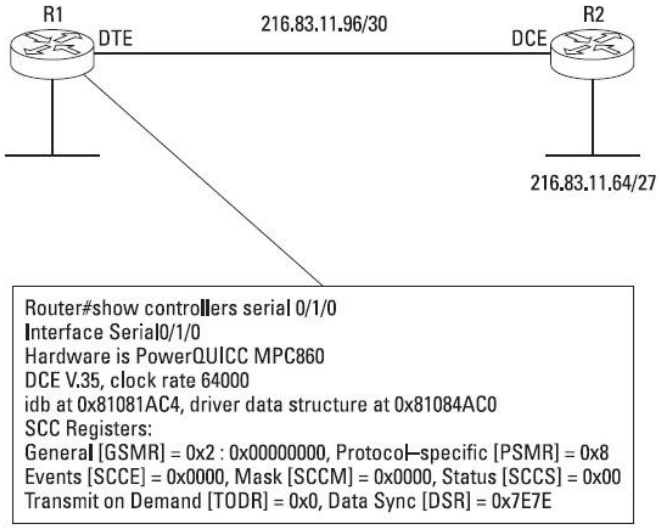
192. You are troubleshooting communication issues between router R1 and R2. The physical cabling is connected as shown in the figure below, as well as the IP address range. You use the show command displayed in the figure to identify the problem. Why are the routers not communicating?



(A) The IP range for the serial ports is incorrect.

(B) The serial interface is using half-duplex.

(C) The serial port has the wrong cable connected.

(D) The serial interface is using the wrong subnet mask.

193. You need to find out if the serial port on your router is acting as the DCE device in the point-to-point link with another router. What command do you need to use?

(A) `show ip interface brief`

(B) `show interfaces`

(C) `show interface serial0/0`

(D) `show controllers serial0/0`

194. You are troubleshooting communication across a serial port and use the show interfaces command to get the output displayed below. What is the problem?

```
Serial0/2/0 is down, line protocol is down
```

(A) The clock rate is not set.

(B) An encapsulation protocol has not been set.

(C) Check the physical connection.

(D) You are using PPP.

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195. Sue, a network administrator in your office, has configured the router with the commands shown below. What is the order of the passwords that would need to be supplied if she connects to the console port on the router to make changes?

```
Router>enable
Router#config term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#enable password cisco
R1(config)#enable secret P@ssw0rd
R1(config)#line vty 0 4
R1(config-line)#password telnetpass
R1(config-line)#
```

(A) P@ssw0rd

(B) cisco

(C) telnetpass / Cisco

(D) telnetpass / P@ssw0rd

(E) telnetpass / Cisco / P@ssw0rd

196. You use the show interfaces command and get the status below on your serial port. Which layer of the OSI model has a problem?

```
Serial0/2/0 is up, line protocol is down
```

(A) Layer-1

(B) Layer-2

(C) Layer-3

(D) Layer-4

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197. Jeff configures the router with the commands shown below. What is the order of the passwords that he would need to use when connecting to the console port of the router to make changes?

```
Router>enable
Router#config term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#enable password cisco
R1(config)#enable secret P@ssw0rd
R1(config)#line con 0
R1(config-line)#password conpass
R1(config-line)#
```

(A) P@ssw0rd

(B) cisco

(C) conpass / Cisco

(D) conpass / P@ssw0rd

(E) conpass / Cisco / P@ssw0rd

198. Using the output below from the show interfaces command, what is most likely the problem?

```
Serial0/2/0 is up, line protocol is down
```

(A) There is no physical link.

(B) A protocol or clock rate has not been set.

(C) You need to configure the router as the DCE device.

(D) You need to configure the router as the DTE device.

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199. Tom configures the router with the commands shown below. What is the order of the passwords that he would need to enter when connecting to the console port of the router to make changes?

```
Router>enable
Router#config term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#enable password cisco
R1(config)#enable secret P@ssw0rd
R1(config)#line con 0
R1(config-line)#password conpass
R1(config-line)#login
```

(A) P@ssw0rd

(B) cisco

(C) conpass / Cisco

(D) conpass / P@ssw0rd

(E) conpass / Cisco / P@ssw0rd